SIDAK DEEP SINGH

singh.sidak1deep@gmail.com • +91 7985553899 • Chennai, India sidakdeep.vercel.app/ •
linkedin.com/in/sidakdeep • github.com/HappyHaru21

SUMMARY

Machine Learning researcher with hands-on experience in sparse neural networks, low-resource NLP, and transformer fine-tuning. Skilled in optimizing model efficiency with PyTorch, LoRA, and quantization, and experienced in experiment tracking with WandB, MLFlow.

EXPERIENCE

Machine Learning Research Assistant 2025-09 - Present
Exprial AI • Cologne, Germany (Remote)

- Collaborating with Experial AI's research pod on applied ML experiments and continual model evaluation.
- Supporting deployment-readiness reviews by translating research insights into actionable engineering next steps
- Driving rapid prototyping cycles for workloads and summarizing experiment findings for stakeholders.

Machine Learning Intern 2025-05 - 2025-07
VIT UNIVERSITY • Chennai, India

- Researched and implemented sparse CNN architectures with orthogonal initialization for imageclassification.
- Developed adaptive per-channel clipping, achieving 84% activation sparsity and 62% accuracy on CIFAR-10.

EDUCATION

B.Tech. CSE (Specialization in A.I. & Robotics) Vellore Institute of Technology • Chennai, India 2023-08 - 2027-06

SKILLS

Python • Pandas • Scikit-learn • PyTorch • Tensorflow • OpenCV • Transformers • Model Training • Neural Networks • Data Preprocessing • LoRA • Quantization • Numpy • Docker • Git • WandB • MLFlow

PROJECTS

OrthoClipCNN (Paper to be published)

Technologies: PyTorch, W&B, CIFAR-10/100

- Designed a CNN with orthogonal weight initialization and per-channel adaptive clipped ReLU activations.
- Achieved ~62% accuracy on CIFAR-10 and ~29.6% on CIFAR-100 with ~84% activation sparsity in early layers.
- Collected activation statistics across layers and tracked all experiments using Weights & Biases.
- https://www.kaggle.com/code/haruki23/ortho-x-clipped-per-channel

Cherokee to English Translation with Gemma-3N

Technologies: PyTorch, Transformers, LoRA, 4-bit quantization, Unsloth, CPU offloading

- Fine-tuned Google's Gemma-3N on ~14k Cherokee-English pairs for low-resource language translation.
- Applied LoRA adapters with 4-bit quantization and gradient checkpointing to fit training on Kaggle T4 GPUs.
- Improved translation quality by shifting from generic instruction prompts to explicit translation prompts.
- https://www.kaggle.com/code/huummm/cherokee-to-english-with-gemma3n

RL for Traffic Optimization (Implementation Phase)

Technologies: MARL, DQN, SUMO

- Designing a MARL framework using DQN agents for adaptive traffic signal control to reduce congestion and improve urban mobility.